Ethan Wang

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EDUCATION

Cornell University, College of Engineering **Expected Dec 2028 | Ithaca, NY**
Bachelor of Science, Mechanical Engineering, GPA: 4.0/4.0
Hunter College High School, GPA: 95.1 / 100 **Graduated June 2025 | New York, NY**

ENGINEERING EXPERIENCE

Cornell Rocketry Team – *Propulsion Team Member* **Nov 2025 – Present**
- Collaborating with 50+ interdisciplinary members to research, design, and build a hybrid propulsion rocket capable of reaching over 10,000 feet.
- Currently developing two motor actuated valves for ethanol and nitrous oxide lines on the 2027 liquid propulsion rocket to withstand sustained 4 ksi pressure, extreme temperature, and heavy loading conditions during launch and flight, all within a 6 inch diameter airframe. Using Solidworks and Matlab fluid simulations to design and source components to ensure a total pressure loss under 10 psi to optimize rocket efficiency.

VEX Robotics Team 955R – *Founder, Captain, Mechanical Team Lead* **Sept 2021 – May 2025**
- Led mechanical design and manufacturing efforts, learning Onshape and Fusion360 to design the robot for annual game scoring requirements, prototype new concepts to optimize lab time, and continuously implement improvements and new parts based on competition results.
- Learned basic 3D printing to fabricate guides and molds for precise aluminum and polycarbonate cutting constrained to a dremmel and band saw.
- Developed leadership skills managing team of six fellow students. Organized meetings around busy schedules and schoolwork, delegated roles based on skills and availability (designers, builders, coders, documentation), and managed projects for consistent improvement and innovation.
- Accolades: 3x World Championships Qualifications (2022, 2024, 2025), 3x Regional Tournament Champion, State Championship Finalist, Innovate Award, Think Award, Top 100 peak robot skills challenge ranking 2025, tied for 3rd in the VEX Worlds 2025 power ranking.

Daewoong Academy – *Backend Intern* **Spring 2024**
- Learned backend web design skills and software with NodeJS, Postman, and MongoDB through building and implementing AI generated SAT question function with a GPT wrapper with fellow interns. Learned LaTeX in Overleaf to format questions and equations for the website.

Teddy Bear Community Nonprofit – *Head Website Designer* **Mar 2023 - Jan 2025**
- Learned Wix to build, maintain, and optimize the website to house 50+ subpages of class info, teacher bios, and public webinars for a non-profit tutoring organization. Oversaw growth from 0 to over 200 students with chapters in 5 different countries and 100+ monthly website viewers.
- Organized a team of volunteers and interns, heading onboarding processes and delegating tasks such as updating class or teacher information, improving the mobile and computer user interface, or creating new pages when new teachers or classes are added.

LEADERSHIP EXPERIENCES

Formosa Association of Student Cultural Ambassadors – *Vice President* **Aug 2023 – Sept 2025**
- Assisted in organizing and promoting events for FASCA, a community service volunteer organization run by the Taiwanese government.
- Monitored member attendance, using Google Sheets with Google Forms integration to efficiently track the service hours of 100+ members.
- Awarded The President's Volunteer Service Award Silver in 2024 for 175+ community service hours within a one year period.

Hunter College High School – *Computer Science Teaching Assistant* **Sept 2024 – June 2025**
- Collaborated with instructor in developing 4 Python lessons taught over the school year covering fundamental concepts like logic, variables, lists, and loops, to eventually computer graphics and interactive elements, culminating in a final project where students build their own game in Python.
- Supported students needing assistance during or out of class, organizing office hours before tests and the final project, as well as one-on-one sessions with students to help improve understanding of Python and general programming concepts, and troubleshooting homework or projects.

CAMPUS INVOLVEMENT

Hunter Boys Varsity Badminton Team – *Captain* **Sept 2021 – Nov 2024**
Badminton Club – *Founder, Co-President* **Sept 2022 – Jun 2025**
Cubing, Puzzles, and Cards Club – *Founder, Co-President* **Sept 2022 – Jun 2024**

TECHNICAL PROFICIENCIES

Design and Simulation: SolidWorks, Onshape, Fusion360, Matlab, Ansys FEA
Machining: Manual Mill, Manual Lathe
Programming: Java, Python, C++, R, Javascript
Communication: Overleaf, Excel, Canva

RELEVANT COURSEWORK

Current: ENGRD 2020: Statics, MATH 2930: Differential Equations for Engineers
Past: ENGRI 1170: Intro to Mechanical Engineering, MATH 1920: Multivariable Calculus for Engineers